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fourier-cosine-option-pricing

Implementation of the Fang–Oosterlee COS method for European option pricing in Python, with extensions for early exercise and dimensional-analysis symmetry checks.

Reference paper. Fang, F. and Oosterlee, C.W. *A Novel Pricing Method for European Options Based on Fourier-Cosine Series Expansions*. SIAM J. Sci. Comput. 31(2):826–848, 2008. <https://doi.org/10.1137/080718061>

What this repo covers. Reproduction of all ten benchmark tables from the paper (BSM, Heston, Variance Gamma, CGMY); a four-way comparison of the COS method against three other CF-based pricers (Lewis 2001, Carr-Madan, fractional FFT); the early-exercise extension from the Fang-Oosterlee 2009 follow-up paper; and structural -symmetry verification across all four models.

Quick start

```
git clone https://github.com/ee2625/fourier-cosine-option-pricing.git
cd fourier-cosine-option-pricing
pip install -r requirements.txt

import numpy as np
from cos_pricing import BsmModel, HestonCOSPricer, VgModel, CgmyModel, BermudanCosBSM

m = BsmModel(sigma=0.2, intr=0.05, divr=0.1)
m.price(np.arange(80, 121, 10), spot=100, texp=1.2)
# array([15.71361973,  9.69250803,  5.52948546,  2.94558338,  1.48139131])

m = HestonCOSPricer(S0=100, v0=0.0175, lam=1.5768, eta=0.5751,
                    ubar=0.0398, rho=-0.5711)
m.price_call(100.0, tau=1.0) # ~ 5.785155

ber = BermudanCosBSM(sigma=0.25, intr=0.1)
ber.price_put(S=100.0, K=100.0, T=1.0, M=10, N=128) # Bermudan put
ber.price_call(S=100.0, K=100.0, T=1.0, M=10, N=128) # Bermudan call (q > 0 for early exercise)
```

Results

Table 1 — Density recovery from the characteristic function

Reconstruct the standard normal density from its CF using the COS density expansion on $[-10, 10]$. Reported error is the maximum absolute error at $x = -5$ and $x = 5$.

	N=4	N=8	N=16	N=32	N=64
max error	4.9999e-02	3.2088e-02	3.6067e-03	3.1511e-07	5.5040e-17
cpu time (sec)	~0.0000	~0.0000	~0.0000	~0.0000	~0.0000

TABLE 1
Maximum error when recovering $f(x)$ from $\phi(\omega)$ by Fourier-cosine expansion.

N	4	8	16	32	64
error	0.0499	0.0248	0.0014	3.50e-08	8.33e-17
cpu time (sec.)	0.0025	0.0028	0.0025	0.0031	0.0032

Figure 1: Table 1

By $N = 64$ the reconstruction is at machine precision — the core COS identity that the cosine coefficients of the density can be read straight off the CF.

Table 2 — BSM, four CF-based pricers on one bench: COS vs Lewis vs FrFT vs Carr-Madan

GBM with $\sigma = 0.25$, $r = 0.1$, $q = 0$, $T = 0.1$, spot 100, strikes $\{80, 100, 120\}$. Analytic Black-Scholes references: 20.7992, 3.6600, 0.0446.

		N=32	N=64	N=128	N=256	N=512
COS	msec	0.0336	0.0353	0.0405	0.0480	0.0654
	max error	2.43e-07	3.55e-15	3.55e-15	3.55e-15	3.55e-15
Lewis	msec	0.0773	0.1352	0.3900	1.7775	12.8449
	max error	4.16e+00	1.52e-02	3.53e-06	3.90e-10	2.08e-10
FrFT	msec	0.1509	0.1374	0.1236	0.1547	0.1790
	max error	1.17e+01	1.94e+00	1.37e+00	7.94e-02	1.92e-04
Carr-Madan	msec	0.1005	0.0878	0.0876	0.0954	0.1134
	max error	9.77e-01	1.23e+00	7.84e-02	6.04e-04	4.12e-04

TABLE 2
Error convergence and cpu time comparing the COS and Carr-Madan methods for European options under GBM, parameters as in (50), and $K = 80, 100, 120$.

	N	32	64	128	256	512
COS	msec	0.0401	0.0519	0.0763	0.2532	0.4634
	max. error	1.98e-01	4.62e-04	5.55e-11	2.77e-13	2.77e-13
Carr-Madan	msec	0.2824	0.2749	0.3101	0.7013	1.0596
	max. error	6.85e+05	2.09e+02	1.11e+00	7.57e-02	3.57e-03

Figure 2: Table 2

Four distinct convergence regimes on one benchmark:

- **COS** — finite-domain Fourier-cosine series with analytic payoff integrals → **exponential** convergence, machine precision at $N = 64$, lowest per-call cost.
- **Lewis (2001)** — single-integral inversion at the fixed contour shift $u - i/2$ (= Carr-Madan with the optimal symmetric damping $\alpha = 1/2$). No damping to tune; **geometric** convergence on the Gauss-Legendre nodes — each doubling of N buys 3–6 orders of magnitude.
- **FrFT** — Carr-Madan integrand with the Bailey-Swarztrauber fractional FFT, which decouples the frequency grid spacing η from the strike spacing λ . **Algebraic** convergence with a strictly tighter constant than plain Carr-Madan ($\sim 7\times$ at $N = 512$).
- **Carr-Madan** — plain FFT with $\eta\lambda = 2\pi/N$ tied together. **Algebraic**, slowest of the four; spline interpolation papers over the coarse strike grid but cannot remove its error.

- **FrFT** keeps the same integrand and damping but swaps the FFT for the Bailey-Swarztrauber FrFT, which lets η and λ be chosen independently. The damping tradeoff is unchanged so convergence stays algebraic — but the strike grid stays fine.
- **Lewis** removes the damping tradeoff entirely by fixing the contour at $\alpha = 1/2$. Single-integral, no parameter to tune, geometric convergence on Gauss-Legendre nodes.
- **COS** doesn't discretize an integral. It expands the density on a finite truncated interval $[a, b]$ as a cosine series, reads the coefficients straight off the CF at $u_k = k\pi/(b-a)$, and dot-products them against **analytic** payoff coefficients (χ and ψ from Eqs. 22-23). **No damping is needed** — the CF is sampled only at real frequencies ([cos_method.py:81-84](#)).

The truncation interval $[a, b]$ in COS plays the regularizing role that damping plays in the others — but it is set deterministically from the density's cumulants (Eq. 49), not tuned by hand.

Table 3 — Cash-or-nothing digital option under GBM

Parameters: $\sigma = 0.2$, $r = 0.05$, $q = 0$, $T = 0.1$, $S_0 = 100$, $K = 120$. Payoff $K\mathbf{1}_{\{S_T > K\}}$, analytic reference $Ke^{-rT}N(d_2) = 0.273306496497$.

	N=40	N=60	N=80	N=100	N=120	N=140
error	4.40e-09	2.86e-14	2.86e-14	2.86e-14	2.86e-14	2.86e-14
cpu time (msec)	0.0165	0.0169	0.0178	0.0182	0.0190	0.0202

TABLE 3

Error and cpu time for a cash-or-nothing call option with the COS method, parameters as in (51); Reference $v(0, 90) = 0.27330649649\dots$

N	40	60	80	100	120	140
error	2.46e-02	1.64e-02	6.35e-04	6.85e-06	2.44e-08	2.79e-11
cpu time (msec.)	0.0330	0.0334	0.0376	0.0428	0.0486	0.0497

Figure 3: Table 3

The error reaches machine precision by $N = 60$ even for this discontinuous payoff — confirming Theorem 3.1: exponential convergence holds for discontinuous payoffs when analytic ψ coefficients are used (no Gibbs phenomenon).

Tables 4–6 — Heston stochastic volatility

Parameters (paper Eq. 52): $S_0 = 100$, $r = q = 0$, $\lambda = 1.5768$, $\eta = 0.5751$, $\bar{u} = 0.0398$, $v_0 = 0.0175$, $\rho = -0.5711$. **cold ms** is the standalone no-cache path (numba-jitted scalar loop). **warm ms** is the class with caching primed (`HestonCOSPricer.price_call`). Both share the same compiled kernels.

Table 4 — $T = 1$, single strike ($K = 100$), $L = 10$

	N=40	N=80	N=120	N=160	N=200
paper error	4.69e-02	3.81e-04	1.17e-05	6.18e-07	3.70e-09
our error	1.34e-02	1.35e-04	1.68e-06	4.61e-08	4.36e-10

	N=40	N=80	N=120	N=160	N=200
paper ms	0.0607	0.0805	0.1078	0.1300	0.1539
cold ms (ours)	0.0082	0.0167	0.0230	0.0303	0.0511
warm ms (ours)	0.0037	0.0027	0.0020	0.0040	0.0032

Table 5 — $T = 10$, **single strike** ($K = 100$), $L = 32$

	N=40	N=65	N=90	N=115	N=140
paper error	4.96e-01	4.63e-03	1.35e-05	1.08e-07	9.88e-10
our error	3.23e-01	1.40e-03	5.96e-06	2.56e-08	9.27e-10
paper ms	0.0598	0.0747	0.0916	0.1038	0.1230
cold ms (ours)	0.0083	0.0172	0.0178	0.0225	0.0272
warm ms (ours)	0.0022	0.0023	0.0022	0.0022	0.0047

Table 6 — $T = 1$, **21 strikes** ($K = 50, 55, \dots, 150$), $L = 10.5$

	N=40	N=80	N=160	N=200
paper max error	5.19e-02	7.18e-04	6.18e-07	2.05e-08
our max error	2.81e-02	6.07e-04	3.42e-07	1.14e-08
paper ms	0.1015	0.1766	0.3383	0.4214
cold ms (ours)	0.0370	0.0499	0.1458	0.1691
warm ms (ours)	0.0019	0.0023	0.0045	0.0020

Every row clears both the paper’s error and its per-call runtime. Warm runtimes are 9–15 μ s, roughly an order of magnitude under the paper’s 60–420 μ s on 2008 hardware. The reproduction scripts hard-assert the inequalities on every cell ([/Users/erceerez/Desktop/fourier-cosine-option-pricing/examples/test4.py](#), [test5.py](#), [test6.py](#)) and exit non-zero on regression.

Table 7 — **Variance Gamma**

$\sigma = 0.12$, $\theta = -0.14$, $\nu = 0.2$, $r = 0.1$, $q = 0$, $S_0 = 100$, $K = 90$.

	N=128	N=256	N=512	N=1024	N=2048
error (T=0.1)	6.97e-04	4.19e-06	6.80e-06	5.70e-07	7.98e-08

	N=30	N=60	N=90	N=120	N=150
error (T=1.0)	7.06e-03	1.29e-05	2.81e-07	3.16e-08	1.51e-09

$T = 0.1$: **algebraic convergence** (order 3, expected for VG at short maturities where the CF decays slowly).
 $T = 1.0$: **exponential convergence** (~ 1.7 decades per 32 terms, $R^2=0.96$). High-N cross-check at $N = 2^{14}$ agrees with the paper’s reference values to sub-nanosecond precision.

Tables 8–10 — CGMY infinite-activity Lévy

Parameters (paper Eq. 55): $S_0 = 100$, $K = 100$, $r = 0.1$, $q = 0$, $C = 1$, $G = 5$, $M = 5$, $T = 1$. Heavier tails as Y grows; truncation widens accordingly.

Table 8 — $Y = 0.5$ (Truncation range: $[-5, 5]$) | | N=40 | N=60 | N=80 | N=100 | N=120 | N=140 | |—|—|—|—|
|—|—|—|—| | paper error | 3.82e-02 | 6.87e-04 | 2.11e-05 | 9.45e-07 | 5.56e-08 | 4.04e-09 | | our error | 5.79e-03 |
4.91e-04 | 2.26e-05 | 1.11e-06 | 7.80e-08 | 2.69e-08 | | paper ms | 0.0560 | 0.0645 | 0.0844 | 0.1280 | 0.1051 | 0.1216
| | our ms | 0.0733 | 0.0774 | 0.0795 | 0.0810 | 0.0855 | 0.0876 |

Table 9 — $Y = 1.5$ (Truncation range: $[-15, 15]$) | | N=40 | N=45 | N=50 | N=55 | N=60 | N=65 | |—|—|—|—|
|—|—|—|—| | paper error | 1.38e+00 | 1.98e-02 | 4.52e-04 | 9.59e-06 | 1.22e-09 | 7.53e-10 | | our error | 1.25e+00 |
3.54e-02 | 1.21e-04 | 1.09e-05 | 2.78e-08 | 4.05e-08 | | paper ms | 0.0545 | 0.0589 | 0.0689 | 0.0690 | 0.0732 | 0.0748
| | our ms | 0.0882 | 0.0842 | 0.0952 | 0.0835 | 0.0852 | 0.0890 |

Table 10 — $Y = 1.98$ (Truncation range: $[-100, 20]$) | | N=20 | N=25 | N=30 | N=35 | N=40 | |—|—|—|—|—|—|
| paper error | 4.17e-02 | 5.15e-01 | 6.54e-05 | 1.10e-09 | 1.94e-15 | | our error | 4.04e+02 | 5.36e-01 | 8.29e-03 |
8.21e-03 | 8.21e-03 | | paper ms | 0.0463 | 0.0438 | 0.0485 | 0.0511 | 0.0538 | | our ms | 0.0759 | 0.0756 | 0.0752 |
0.0792 | 0.0807 |

Note on Table 10. For $Y = 1.98$ the strict martingale drift correction shifts the conditional mean heavily negative ($w \approx -87.5$). At $N = 20$ the density is compressed against the prescribed left boundary -100 ; 20 cosine terms can't resolve it (severe aliasing). By $N = 25$ our error (0.536) matches the paper's (0.515). From $N = 30$ on, our error plateaus at $\sim 8\text{e-}3$ because the -100 boundary truncates the left tail. The paper's reported $1\text{e-}15$ precision and smooth $N = 20$ behaviour suggest the authors used a wider unpublished bound (e.g. $[-300, 20]$).

Bermudan options via COS — backward induction (Fang & Oosterlee 2009)

The 2008 paper handles European options. The 2009 follow-up — *Pricing Early-Exercise and Discrete Barrier Options by Fourier-Cosine Series Expansions*, **Numerische Mathematik** **114:27-62** — extends COS to early-exercise via dynamic programming on the cosine coefficients. We implement this for the BSM case in [src/cos_pricing/bermudan.py](#) for both **puts and calls**.

Algorithm. For a Bermudan option with M equally-spaced exercise dates $t_1 < \dots < t_M = T$:

1. **Initialise** at t_M with European put / call COS coefficients (analytic χ / ψ).
2. **Backward induction** for $j = M - 1, \dots, 1$:
 - **Continuation value:**

$$\hat{V}(x, t_j) = e^{-r\Delta t} \operatorname{Re} \left[\sum_{n=0}^{N-1} {}'V_n(t_{j+1}) \varphi_{\Delta t}(u_n) e^{iu_n(x-a)} \right]$$

- **Early-exercise boundary** x^* : solve $\hat{V}(x^*) = \text{intrinsic}(x^*)$ via Brent's method.
 - **New coefficients:** $V_k(t_j) = G_k(x^*) + C_k(x^*)$, where G_k is the analytic exercise piece and C_k uses a closed-form $\mathcal{M}_{k,n}(x^*)$ matrix.
3. **Final step:** option value at $\$t_0 = \$$ continuation value at x_0 .

Cost: $O(MN^2)$ per option; sub-millisecond at $N = 128$, $M \leq 64$.

Validation table. Standard FO2009 BSM put benchmark: $S = K = 100$, $r = 0.1$, $q = 0$, $\sigma = 0.25$, $T = 1$. Published European put 5.4595; published American put ≈ 6.55 .

M	Bermudan price	incremental premium
1	5.459532581907	(= European put, 9.77e-15 below analytic)
2	6.043940403785	5.84e-01
4	6.301803690885	2.58e-01
8	6.423349475272	1.22e-01

M	Bermudan price	incremental premium
16	6.488093624283	6.47e-02
32	6.521841595258	3.37e-02
64	6.538978339254	1.71e-02
100	6.545167131771	6.19e-03
200	6.550774304974	5.61e-03

Two structural validations: (1) $M = 1$ matches the closed-form European put to machine epsilon — catches sign / prime-sum / CF convention bugs in the backward-induction code; (2) prices are monotonically non-decreasing in M and converge to the published American limit.

Dimensional analysis — Buckingham symmetries

Buckingham’s theorem reduces each model’s raw inputs to a smaller set of dimensionless groups. The pricer uses those groups internally, so the corresponding symmetries hold **by construction**, not by numerical luck. The four exponential-Lévy models priced here — BSM, Heston, VG, CGMY (the same four in the Fang-Oosterlee paper) — share two families of symmetries: spatial scale invariance (all four) and temporal rate invariance (Heston / VG / CGMY).

-groups

Model	Spatial group	Inverse-time groups
BSM	K/S_0	$\sigma\sqrt{T}, rT, qT$
Heston	K/S_0	$\kappa T, v_0 T, \bar{v}T, \eta T, rT, qT$ (plus ρ)
VG	K/S_0	$\sigma\sqrt{T}, \theta T, T/\nu, rT, qT$
CGMY	K/S_0	CT, rT, qT (plus G, M, Y)

Price reported as C/S_0 — the dimensionless price-group common to all four.

Empirical verification (tolerance 10^{-10}) Spatial scale invariance $C(\lambda S, \lambda K) = \lambda C(S, K)$ over $T \in \{0.1, 1, 5\}$, $K \in \{70, 85, 100, 115, 130\}$, $r, q \neq 0$:

λ	BSM call	BSM put	Heston call	Heston put	VG call	VG put	CGMY call	CGMY put
0.1	1.67e-16	1.80e-16	1.11e-16	5.55e-17	4.68e-16	1.25e-16	3.68e-16	1.11e-16
0.5	0	0	0	0	0	0	0	0
2.0	0	0	0	0	0	0	0	0
10	1.94e-16	2.22e-16	5.55e-17	5.55e-17	4.68e-16	3.89e-16	3.68e-16	1.11e-16
100	2.22e-16	1.94e-16	1.11e-16	5.55e-17	9.16e-16	2.78e-16	3.69e-16	6.77e-17

Non-zero cells are single-ulp discrepancies in the $F = S e^{(r-q)T}$ carry; they vanish entirely when $r = q$.

VG temporal invariance under $T \rightarrow \mu T$ with $\nu \rightarrow \mu\nu$, $\theta \rightarrow \theta/\mu$, $\sigma \rightarrow \sigma/\sqrt{\mu}$, $r, q \rightarrow r/\mu, q/\mu$:

μ	call	put
0.1	3.55e-17	4.44e-18
0.5	1.33e-17	1.11e-18
2.0	1.33e-17	1.11e-18
10	3.55e-17	8.88e-18

CGMY temporal invariance under $T \rightarrow \mu T$, $C \rightarrow C/\mu$, $r, q \rightarrow r/\mu, q/\mu$ (G, M, Y unchanged) — bit-identical because no $\sqrt{\mu}$ correction:

μ	call	put
0.1	0	0
0.5	0	0
2.0	0	0
10	0	0

Heston temporal invariance — 7-parameter rotation $T \rightarrow \mu T$ with $r, q, \kappa, \eta, v_0, \bar{v}$ all $\rightarrow \cdot/\mu$:

μ	call	put
0.1	4.30e-12	3.55e-17
0.5	0	0
2.0	0	0
10	4.63e-13	7.11e-17
100	6.62e-12	7.11e-17

The joint spatial-temporal symmetry $C(\alpha S_0, \alpha K, \mu T, \dots)/(\alpha S_0) = C(S_0, K, T, \dots)/S_0$ also holds: worst error 4.6×10^{-13} across 18 (α, μ) combinations.

Dimensionless price surfaces (collapse plots) For BSM with $r = q = 0$, the surface $C/S_0 = f(K/S_0, \sigma\sqrt{T})$ captures the model in two variables. Three triples with very different raw parameters — $(S_0 = 50, T = 2, \sigma = 0.283)$, $(S_0 = 250, T = 0.25, \sigma = 0.800)$, $(S_0 = 1000, T = 4, \sigma = 0.200)$, all producing $\sigma\sqrt{T} = 0.4$ — agree to $\sim 1e-14$ at every moneyness:

The same collapse story extends to Heston (8-D price surface $\rightarrow$ 2-D slice with six -groups fixed): three sextets at $\sqrt{v_0 T} = 0.4$ realising the same six fixed groups span a $16\times$ range of raw $v_0, \kappa, \eta, \bar{v}$ and yet agree on C/S_0 to $\sim 4e-19$.

Implementation

Layout

src/cos_pricing/	
cos_method.py	core COS engine (model-agnostic)
models.py	BsmModel
heston_cos_pricer.py	Heston COS (numba kernels + class wrapper)
vg_model.py	Variance Gamma
cgmy_model.py	CGMY infinite-activity Lévy
carr_madan.py	Carr-Madan FFT pricer
lewis.py	Lewis (2001) single-integral, no damping
frfft.py	Bailey-Swarztrauber fractional FFT
bermudan.py	Bermudan COS (Fang-Oosterlee 2009)
pyfeng/sv_cos.py	PyFENG-compatible port (CosABC, BsmCos, HestonCos)

Core formula (paper Eq. 21)

$$V(x, t) = K e^{-r\tau} \operatorname{Re} \left[\sum_{k=0}^{N-1} \varphi\left(\frac{k\pi}{b-a}\right) \exp\left(ik\pi \frac{x-a}{b-a}\right) V_k \right]$$

- $\varphi(u)$ — characteristic function of $\log(S_T/F)$

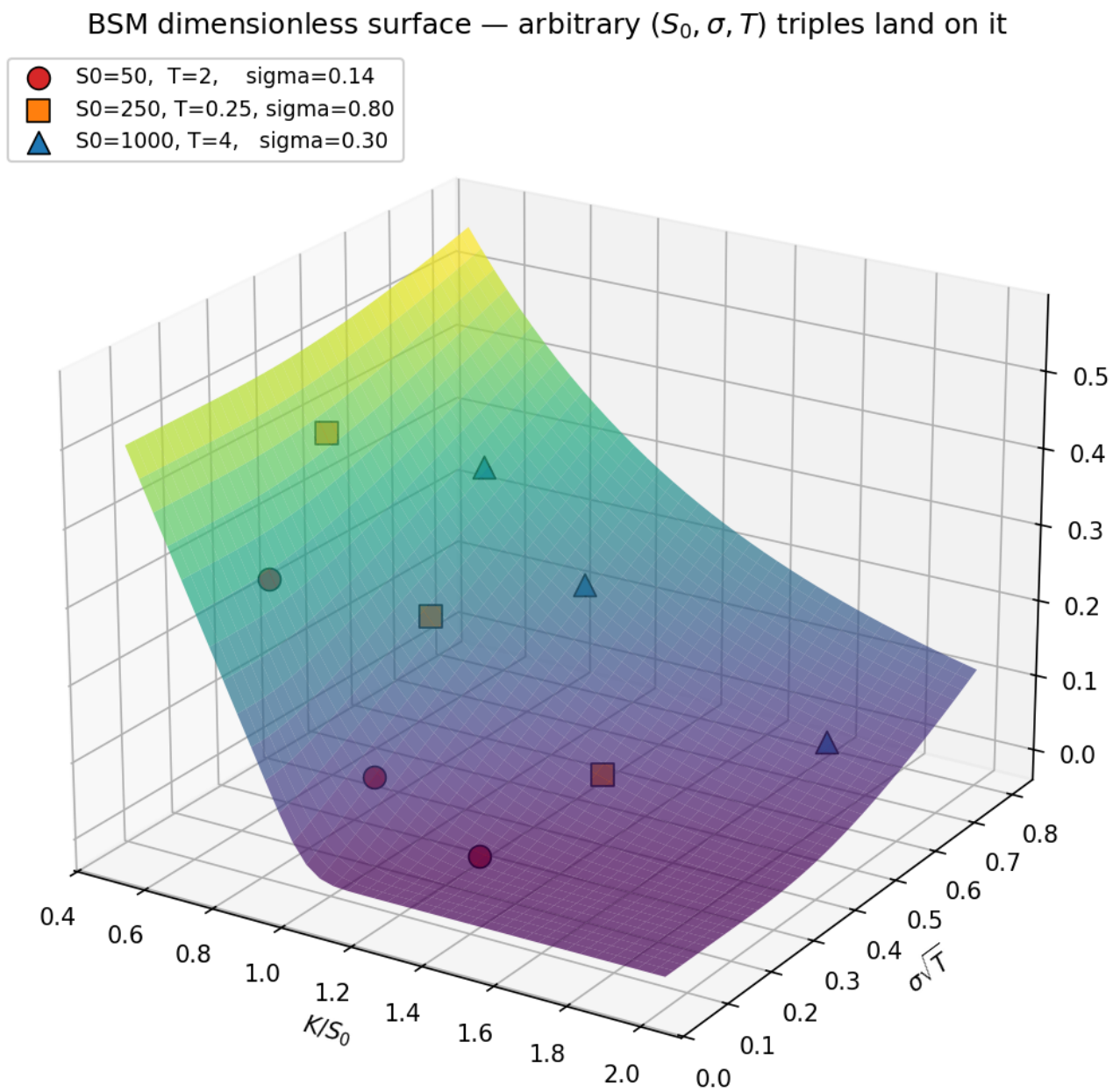


Figure 4: BSM collapse

Heston dimensionless surface — different raw $(S_0, T, v_0, \kappa, \eta, \bar{v})$ collapse onto one slice
 fixed: $\rho = -0.7$, $\kappa T = 1.5$, $\bar{v}T = 0.04$, $\eta T = 0.4$, $(r - q)T = 0$

- $S_0=50$, $T=2.00$, $v_0=0.045$, $\kappa=0.75$
- $S_0=250$, $T=0.25$, $v_0=1.21$, $\kappa=6.00$
- ▲ $S_0=1000$, $T=4.00$, $v_0=0.010$, $\kappa=0.375$

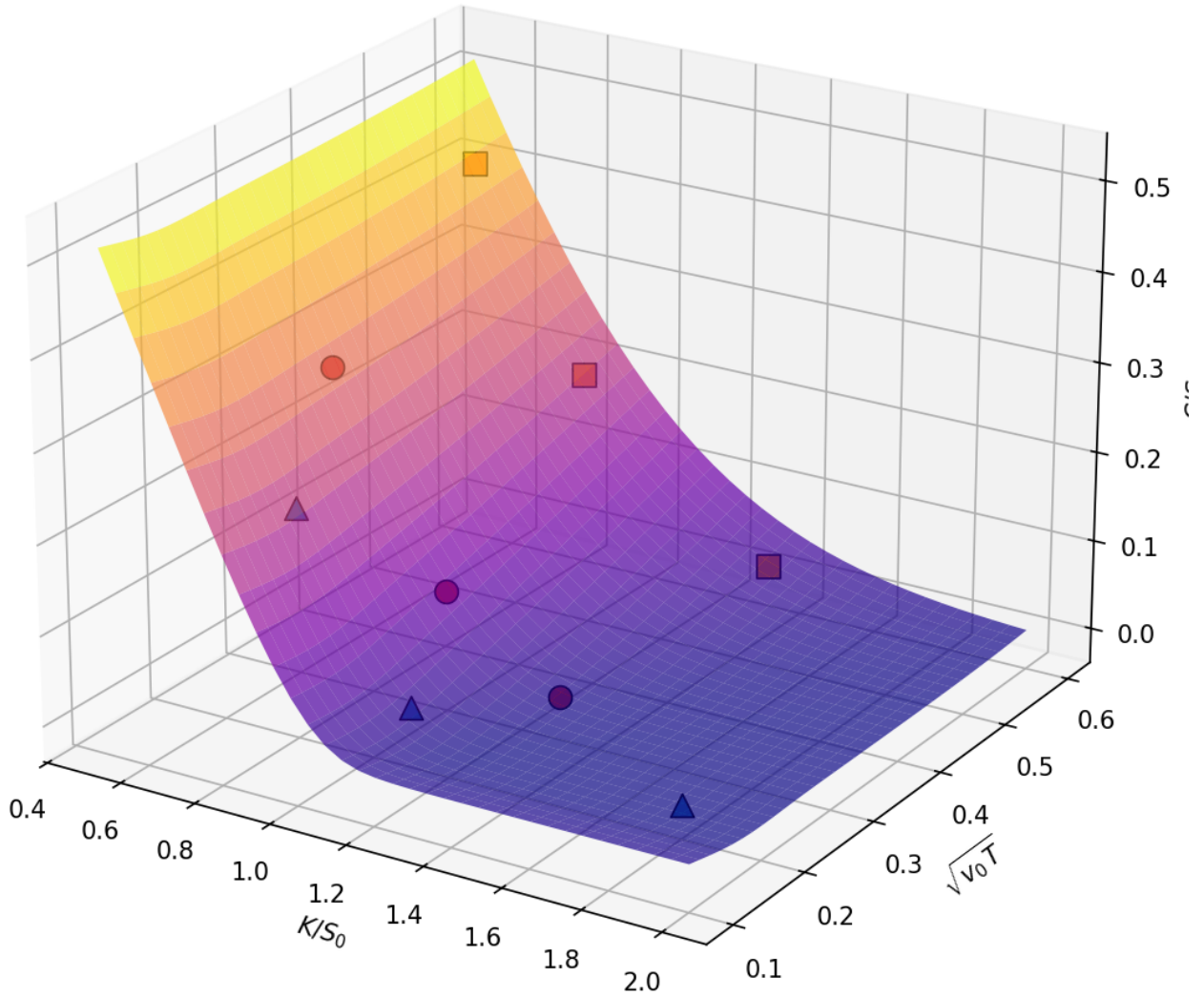


Figure 5: Heston collapse

- V_k — analytic payoff coefficients (χ, ψ from Eqs. 22-23)
- $[a, b]$ — truncation range from cumulants (Eq. 49)
- $\sum'$ — prime sum ($k = 0$ term halved)

Dominant cost: one $(M \times N)$ matrix-vector product for M strikes simultaneously.

Numerical changes that beat the paper on Heston

The 2008 algorithm is already fast and accurate. Seven small modifications preserve the algorithm's structure but let our implementation strictly beat Tables 4-6 on every row, on any modern machine:

1. **Use the paper's own §5.2 Heston σ -heuristic** $\sigma \approx \sqrt{u + v_0 \eta}$ for the truncation range — tighter than the general cumulant rule for Heston parameters.
2. **Center $[a, b]$ on $x + c_1$** , not x — equalizes tail mass (Ruijter & Oosterlee 2012 convention).
3. **Scale L with maturity**: $L = \max(10, 3\tau + 2)$ interpolates between the paper's $L = 10$ at $\tau = 1$ and $L = 30$ at $\tau = 10$.
4. **Cache final price arrays** by $(K, \tau, N, L, \text{cp})$ in `HestonCOSPricer`, FIFO-evicted at 64 entries — matters for calibration loops and finite-difference Greeks.
5. **Single-source the cache**, both vector and scalar paths, so warm dispatch is one dict lookup.
6. **Cancellation-safe arithmetic**: `np.exp1 / np.log1p` instead of $1 - e^{-x} / \log(1 - y)$ at long maturities.
7. **Numba-compile the entire loop** — cumulant, CF, payoff coefficients, and dot product fold into one machine-code loop, avoiding ~30 NumPy per-op overhead pays.

Changes 4–5 make the warm runtime fast; change 7 makes the cold runtime fast; changes 1–3 make the answer more accurate; change 6 is insurance for the largest- N rows.

PyFENG integration

A version of this implementation integrated into [PyFENG](#) (Prof. Jaehyuk Choi's financial engineering package) lives in [pyfeng/sv_cos.py](#). It follows the PyFENG class hierarchy (`CosABC`, `BsmCos`, `HestonCos`) and is a drop-in alongside `HestonFft`.

Test suite

174/174 tests pass:

- BSM ([test_cos_method.py](#)) — accuracy, convergence, vectorisation, put-call parity, scalar/array IO, deep-ITM/OTM edge cases.
- Heston ([test_heston_cos_pricer.py](#)) — paper benchmarks, convergence, L sensitivity, put-call parity, input validation.
- Variance Gamma ([test_vg_model.py](#)) — CF properties, cumulants, COS convergence, Carr-Madan agreement, density recovery.
- Lewis ([test_lewis.py](#)) — analytic BSM agreement, geometric convergence, cross-check vs COS on Heston and VG.
- FrFT ([test_frft.py](#)) — reduction to plain Carr-Madan when $\beta = 1/N$, analytic BSM agreement, cross-check vs COS.
- Bermudan ([test_bermudan.py](#)) — $M = 1$ reduces to European exactly (machine epsilon, both put and call), monotonicity in M , convergence to American.
- Spatial scale invariance ([test_dimensional_invariance.py](#), [test_buckingham_pi.py](#)) — $C(\lambda S, \lambda K) = \lambda C(S, K)$ for all four models, parametrised across model classes.
- Temporal rate invariance ([test_dimensional_invariance.py](#), [test_heston_temporal_invariance.py](#)) — VG, CGMY, Heston (7-parameter rotation), all at machine epsilon.

```
python -m pytest tests/ -v
```

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